Train the agent, not just the model

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THE VOCABULARY

An agent is a model plus a harness

Claude Code · Codex · OpenCode · Pi · Mini-SWE-Agent

🔁 runs the loop

it calls the model again and again, and decides when to stop

🛠️ owns the tools

it decides which tools exist, and it is the one that runs every call

📝 writes the context

system prompt, history, compaction: everything the model reads

🧩 reads the reply

in its own format, and rejects it if it does not fit

Same weights, different harness: LFM2.5-2.6B solves **62%** of the tasks in Mini-SWE-Agent and **33%** in Claude Code.

Source: The ultimate guide to multi-harness RL

THE MAP

How an agent gets trained in 2026

Post-training is where a pretrained model becomes an agent.

WHAT THE MODEL LEARNS FROM

imitate

SFT on agent traces

a demonstrator: the agent's past sessions

SFTTrainer

compress

Distillation

a teacher: its probabilities on every token

DistillationTrainer

reward

RL with verifiable rewards

a check you can run: did the tests pass?

GRPOTrainer

WHERE IT ACTS WHILE IT LEARNS

act

RL in environments

a sandbox with tools, in a loop the trainer runs

GRPOTrainer + OpenEnv

harness

RL through the harness

the real agent, running its own loop

AsyncGRPOTrainer + OpenEnv

The first three change the training signal, the last two move training into a loop where the model acts. **Mix and match: combine them, change the order, and skip what your data cannot support.**

THE STACK

Two open projects for every method

TRL

TRAINS THE MODEL

Hugging Face’s post-training library. One trainer per method, with the same shape for all of them.

- SFTTrainer
- DistillationTrainer
- GRPOTrainer
- AsyncGRPOTrainer

hf.co/docs/trl

OpenEnv

SERVES THE ENVIRONMENTS

One interface for RL environments, reset and step behind an HTTP server, each one a Space you can pull from the Hub.

- coding_env
- Harbor tasks
- browsers
- games
- simulators

and real coding agents inside them:

hf.co/docs/openenv

Both run on PyTorch.

IMITATE

SFT on agent traces

PRE-RECORDED TRACE (ONE WHOLE AGENT SESSION, SAVED AS DATA) — static dataset, nothing runs while training

1. USER fix the failing test in utils.py	2. AGENT read("utils.py")	3. TOOL def parse(s): ...	4. AGENT edit("utils.py", ...)
5. TOOL pytest: 1 failed	6. AGENT edit("utils.py", ...)	7. TOOL pytest: 3 passed	8. AGENT Fixed the off-by-one in parse()

☑ in the loss ☐ context only

Messages, tool calls and tool results, in the chat format the model already reads.

With **assistant_only_loss** the model learns the agent's turns and only reads the rest.

Claude Code, Codex and Pi traces can be shared on the Hub, with a trace viewer.

> The model only gets as good as the traces it copies, and good traces rarely show how to recover from a mistake.

Source: Agent traces on the Hub (docs)

IMITATE

SFT in TRL

The traces become chat messages, with their tool calls, and **SFTTrainer** trains on them.

sft.py PYTHON

```python
from huggingface_hub import snapshot_download
from trl import SFTConfig, SFTTrainer

# raw Pi sessions (JSONL), as the agent recorded them
path = snapshot_download("badlogicgames/pi-mono", repo_type="dataset")

# Pi's own format -> chat messages, keeping tool_calls and tool turns
dataset = sessions_to_messages(path)

trainer = SFTTrainer(
    model="Qwen/Qwen3-0.6B",
    train_dataset=dataset,
    args=SFTConfig(assistant_only_loss=True),
)
trainer.train()
```

→ Only because pi-mono is raw Pi JSONL. Traces already in chat format load with **load_dataset** and skip this.

Training Agents #1: SFT SFT docs pi-mono traces Episode 1 scripts

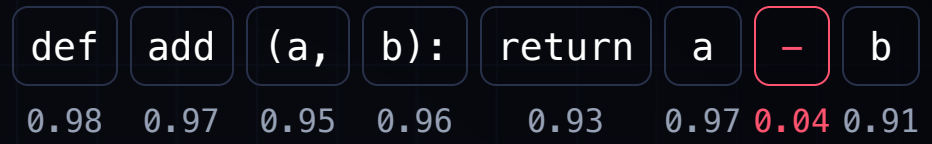
COMPRESS

Distillation: learn from a teacher

`return a _` next token? (illustrative)



ON-POLICY: THE TEACHER GRADES THE STUDENT'S TOKENS



The teacher puts 0.04 on −, so the student is corrected at the exact token where it went wrong.

A teacher gives a probability for every candidate token, where SFT gives only the right one.

On-policy, the student learns on text it wrote itself, so it sees its own mistakes.

The teacher: a bigger model, RL experts merged into one (MOPD), or **the model itself with a hint** (self-distillation).

> Teacher and student need the same tokenizer, and the student is limited by how good the teacher is.

Source: Distillation in 2026

COMPRESS

Distillation in TRL

The student generates from the prompts and the teacher scores every token. On-policy by default.

distill.py PYTHON

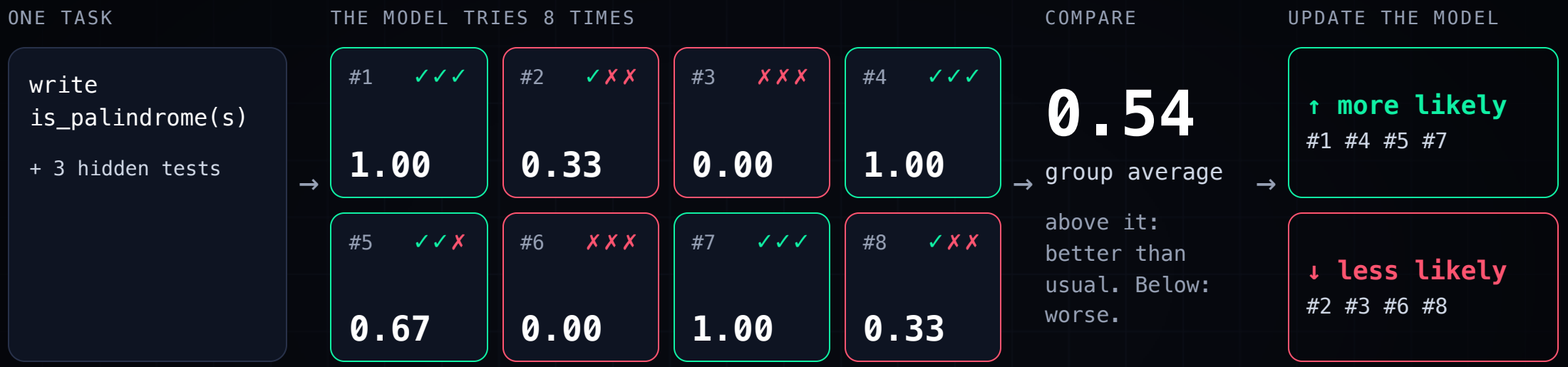
```python
from trl import DistillationTrainer

trainer = DistillationTrainer(
    model="Qwen/Qwen3-0.6B",          # student
    teacher_model="Qwen/Qwen3-4B",    # same tokenizer
    train_dataset=prompts,            # prompts only
)
trainer.train()
```

Training Agents #2: distillation Distillation in 2026 Distillation docs

REWARD

GRPO: learn from a check you can run



EACH ✓ IS ONE HIDDEN TEST PASSED, AND THE FRACTION PASSED IS THE REWARD (ILLUSTRATIVE NUMBERS)

The group is its own baseline, so there is no separate value model to train.

Tests, a compiler or a known answer all work as the reward.

Nobody shows the model the answer, so no demonstrator caps the result.

REWARD

GRPO in TRL

The reward is a plain Python function over the completions.

grpo.py PYTHON

```python
from trl import GRPOConfig, GRPOTrainer

def tests_pass(completions, test_list, **kwargs):
    # fraction of the task's asserts the code passes
    return [run_asserts(c[0]["content"], t)
            for c, t in zip(completions, test_list)]

trainer = GRPOTrainer(
    model="Qwen/Qwen3-0.6B",
    reward_funcs=tests_pass,
    train_dataset=mbpp,
    args=GRPOConfig(num_generations=8),  # the group
)
trainer.train()
```

Training Agents #3: RL Agentic RL in 2026 GRPO docs

ACT

RL inside an environment

GRPOTRAINER RUNS THIS LOOP, ONE SANDBOX PER ATTEMPT (ILLUSTRATIVE)



☐ model turns, trained on ☐ the OpenEnv sandbox answers

OpenEnv gives every environment one contract (reset, step), served as a Hub Space.

The reward moves into the sandbox, and comes after many turns of tool use.

ACT

Environments in TRL + OpenEnv

Same **GRPOTrainer** as before. The class's methods become tools, **get_reward** is the reward.

grpo_env.py PYTHON

```python
from trl import GRPOConfig, GRPOTrainer
from coding_env import CodeAction, CodingEnv  # OpenEnv client

class MBPPEnv:
    def reset(self, tests=None, **kwargs): ...  # fresh sandbox
    def get_reward(self) -> float: ...          # hidden tests
    def run_python(self, code: str) -> str:     # a tool
        r = self.client.step(CodeAction(code=code))
        return r.observation.stdout + r.observation.stderr

trainer = GRPOTrainer(
    model="Qwen/Qwen3-1.7B",
    train_dataset=mbpp_tasks,
    environment_factory=MBPPEnv,
    args=GRPOConfig(num_generations=8),
)
trainer.train()
```

HARNESS

Frontier labs now train coding models across several real harnesses

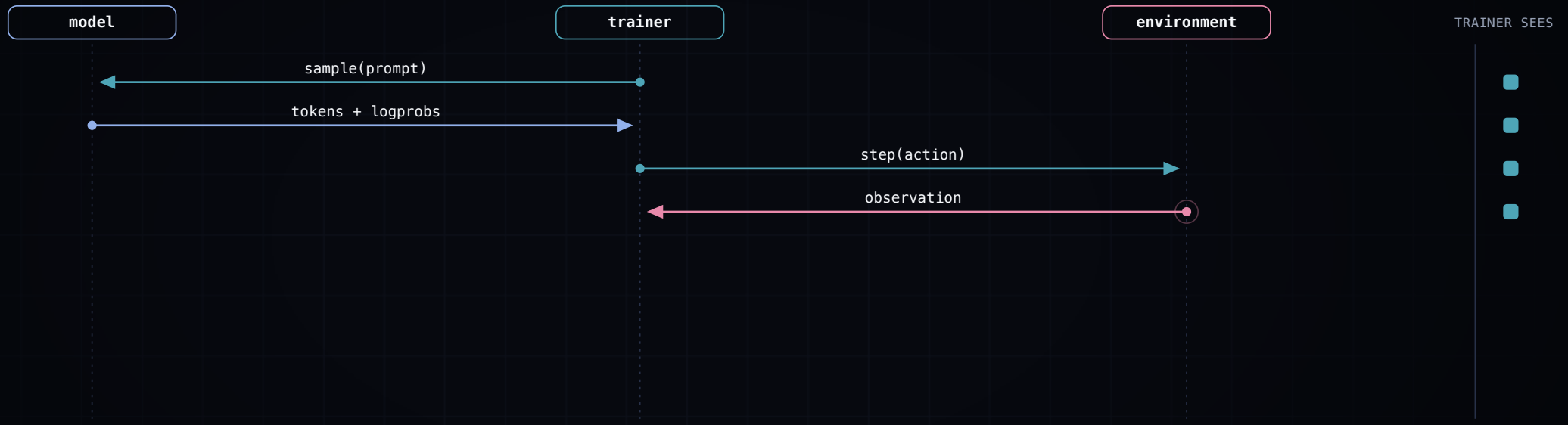
Kimi K3	builds its harnesses from modules, trains across them
Qwen3-Coder-Next	agentic coding data from six harnesses
MiMo-V2.6	a pool of task-adapted mini-harnesses
Poolside Laguna	1.3B tokens of OpenHands, OpenCode, Mini-SWE-Agent
Liquid LFM2.5	a harness picked at random per task

Source: The ultimate guide to multi-harness RL

HARNESS

Who drives the loop: the trainer or the harness

White box | Black box



4/4
EXCHANGES THE TRAINER STARTS

8/8
EVENTS THE TRAINER OBSERVES

THE TRAINER DRIVES IT
WHO OWNS THE LOOP

ARROW COLOUR IS WHOEVER MADE THE CALL

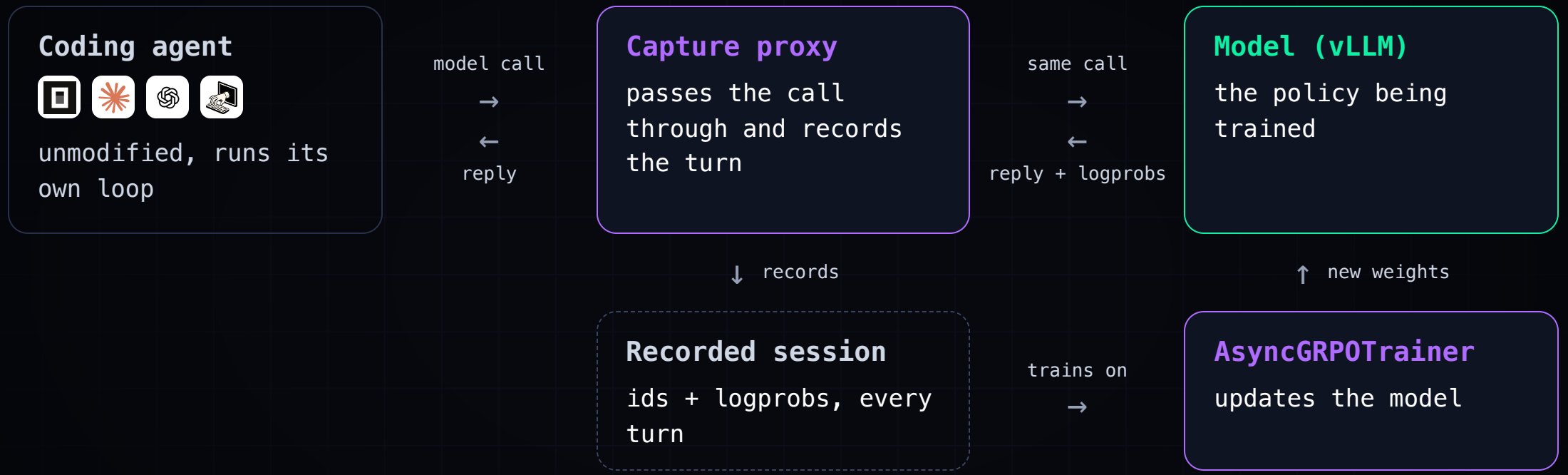
model — trainer — environment — dashed: the trainer is not there

The trainer starts every exchange, so it sees the whole episode.

The environment we just saw was the white box: our trainer called the model and the tools. A harness is the black box. It calls the model itself.

HARNESS

The capture proxy



The harness talks to what it thinks is a normal model API. The proxy forwards each call to vLLM and records the token ids and logprobs of every turn, which is what GRPO needs to train.

HARNESS

In TRL + OpenEnv

multi_harness.py PYTHON

```python
from harbor_env import HarborEnv  # OpenEnv client
from trl.experimental.async_grpo import AsyncGRPOTrainer
from trl.experimental.async_grpo.openenv_harness import HarnessRolloutWorker

env = HarborEnv.from_hub(
    "FineEnvs/smoldataenv-multi-harness-harbor",  # tasks and sandboxes
    harness=random.choice(HARNESSES),  # picked per rollout
)

worker = HarnessRolloutWorker(
    harness_session_factory=env.session_factory,
    harness_adapter=None,  # the agent owns its loop
    rollout_reward_fn=reward,  # correctness and tool calls
)

AsyncGRPOTrainer(model=..., rollout_worker=worker).train()  # keeps generating while it trains
```

The harnesses come from Harbor, and OpenEnv runs them. Swapping the harness is **one argument**.

Simplified for the slide. Full training script Multi-harness RL

RECAP

Which one to reach for

You have traces of the agent you want	**SFT**
You have a stronger model with your tokenizer	**Distillation**
You can check the answer	**GRPO**
The task takes many steps with tools	**GRPO + OpenEnv**
Users run it inside a harness	**Train through the harness**

They combine. DeepSeek-V4 runs SFT then GRPO per domain, then merges the experts with on-policy distillation.

OVER TO YOU

Go deeper

TRAINING AGENTS, LIVE	READ	RUN	CODE
#1 SFT on agent traces	Distillation in 2026	SFT on tool calls	TRL
#2 Distillation	Agentic RL in 2026	Distillation	OpenEnv
#3 RL with GRPO	RL environments in 2026	GRPO + OpenEnv	TRL docs
#4 RL environments	Multi-harness RL	AsyncGRPO + Harbor	OpenEnv docs
The whole playlist	Agent glossary	Multi-harness models	Environments on the Hub

The deck is at hf.co/spaces/sergiopaniego/train-the-agent-slides

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THESE SLIDES